

Business Requirement Document

Order Management Module

ShopSphere E-Commerce Platform — ShopSphere Platform Modernization Program (FY2026)

Document Title	Business Requirement Document — Order Management Module
Document ID	BRD-EC-ORM-005
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Author	Neha Kapoor, Business Analyst — Order Operations
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1. Document Information

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2. Executive Summary

ShopSphere Technologies Pvt. Ltd. operates the ShopSphere E-Commerce Platform, a B2C marketplace serving approximately 2 million monthly active users across India, UAE, and Southeast Asia. As part of the ShopSphere Platform Modernization Program for fiscal year 2026, this Business Requirement Document defines the requirements for the Order Management module — the system of record that governs the end-to-end lifecycle of a customer order from the moment it is placed to the moment it is closed. The current order management capability is fragmented across legacy order-processing scripts and manual warehouse coordination, resulting in inconsistent order status visibility, delayed exception handling when stock is unavailable post-order, and limited ability to split large or multi-warehouse orders efficiently. This document specifies the business, functional, and non-functional requirements needed to establish a single, authoritative source of truth for order status across web, mobile, and customer service channels; to enforce SLA-driven order-to-ship timelines; and to support automated order splitting and logistics handoff across ShopSphere's fulfillment network. The scope covers order lifecycle orchestration, cancellation handling prior to shipment, partial shipment and warehouse-level order splitting, and handoff to logistics partners, while explicitly excluding last-mile route optimization, warehouse picking execution, full returns/refund processing, and marketplace seller order routing, each of which is addressed in dedicated modules or separate BRDs.

3. Business Objective

The primary business objective of the Order Management module is to deliver a resilient, transparent, and auditable order processing capability that reduces order fulfillment delays, minimizes customer-facing order status discrepancies, and improves operational efficiency across ShopSphere's multi-warehouse fulfillment network. Specifically, the module aims to: (a) establish a single source of truth for order status that is consistently reflected across the ShopSphere web storefront, mobile app, customer service tools, and third-party logistics tracking pages; (b) reduce order cancellation and customer complaint volume caused by post-order stock unavailability through proactive exception handling and automated order splitting; (c) enforce measurable SLA adherence for order-to-ship time to support ShopSphere's competitive delivery promise; and (d) provide Order Operations, Finance, and Customer Service teams with real-time, accurate order data to support decision-making, revenue recognition in SAP S/4HANA, and customer support case resolution in Salesforce CRM.

4. Project Scope

In Scope

- End-to-end order lifecycle state management: Placed → Confirmed → Packed → Shipped → Out for Delivery → Delivered → Closed.
- Real-time order status tracking and synchronization across web, mobile app, and customer service channels.
- Customer-facing order status notifications via email and SMS at each significant lifecycle transition.
- Pre-shipment order cancellation, including customer-initiated and system-initiated (stock shortfall) cancellations.
- Partial shipment and order splitting logic across multiple warehouses when a single order cannot be fully fulfilled from one location.
- Logistics partner assignment, shipment creation, and handoff to SwiftShip Logistics (domestic India) and Continental Freight Partners (UAE/Southeast Asia cross-border).
- Returns-initiation handoff — capturing the customer's return request against a delivered order and passing it to the Returns & Refunds module.
- Exception handling for out-of-stock conditions detected after order confirmation.

Out of Scope

- Last-mile delivery route optimization and delivery agent scheduling (owned by logistics partners' own systems).
- Warehouse picking and packing execution processes internal to NetSuite WMS.
- Full returns and refund processing, including refund computation and disbursement (covered under a separate Returns & Refunds BRD).
- Marketplace seller order routing and seller-fulfilled order workflows.
- Payment authorization and capture logic (owned by the Payments module and PayEdge Payment Gateway integration).

5. Stakeholders

Name	Role	Department	Responsibility
Rajesh Mehta	VP of E-Commerce	Business	Executive Sponsor
Ananya Sharma	Product Owner, ShopSphere Platform	Product	Prioritization & sign-off
Neha Kapoor	Business Analyst, Order Operations	Business Analysis	Requirements owner
Sarah Chen	Enterprise IT Architect	IT/Architecture	Technical feasibility
Michael Torres	QA Lead	Quality Assurance	Test strategy & sign-off
David Kim	DevOps Lead	Engineering/Infra	Deployment & monitoring
Fatima Al-Sayed	Compliance & Security Officer	Risk & Compliance	Regulatory compliance

6. Business Requirements

ID	Business Requirement
BR-001	ShopSphere must maintain a single, authoritative order status that is consistent across the web storefront, mobile app, customer service console, and logistics tracking pages at all times.
BR-002	ShopSphere must be able to automatically split an order across multiple warehouses when a single warehouse cannot fulfill the complete order, without requiring manual intervention from Order Operations staff.
BR-003	ShopSphere must allow customers to cancel an order prior to shipment through self-service channels, reducing dependency on customer support calls.
BR-004	ShopSphere must detect and gracefully handle situations where inventory becomes unavailable after order confirmation but before shipment, minimizing customer dissatisfaction.
BR-005	ShopSphere must enforce and monitor order-to-ship SLA targets to support the platform's delivery promise commitments to customers.
BR-006	ShopSphere must synchronize confirmed order data with SAP S/4HANA in near real time to support accurate revenue recognition and order-to-cash processes.
BR-007	ShopSphere must provide Customer Service representatives with complete, real-time order history and status visibility within Salesforce CRM to resolve customer inquiries efficiently.

7. Functional Requirements

Order Lifecycle

- **FR-001** The system shall create a new order record in "Placed" status immediately upon successful payment authorization from the PayEdge Payment Gateway or successful COD order validation.
- **FR-002** The system shall transition an order to "Confirmed" status only after inventory reservation is verified as successful in the Inventory Management System (IMS) and NetSuite WMS.
- **FR-003** The system shall update order status to "Packed" upon receipt of a packing-complete confirmation event from NetSuite WMS for all line items in the order or order segment.
- **FR-004** The system shall update order status to "Shipped" upon receipt of a shipment manifest confirmation from the assigned logistics partner, capturing the tracking number and carrier reference.
- **FR-005** The system shall update order status to "Out for Delivery" and "Delivered" based on tracking status webhooks received from SwiftShip Logistics or Continental Freight Partners.
- **FR-006** The system shall automatically transition an order to "Closed" status seven calendar days after the "Delivered" status is recorded, provided no return has been initiated within that window.
- **FR-007** The system shall broadcast every order status transition to the Notification Service to trigger customer-facing email (via SendGrid Email API) and SMS (via Twilio SMS Gateway) alerts.

Cancellations

- **FR-008** The system shall allow a customer to cancel an order or an individual line item through self-service channels at any point prior to the order or item entering "Packed" status.
- **FR-009** The system shall automatically release reserved inventory in the IMS immediately upon order cancellation confirmation.
- **FR-010** The system shall trigger a system-initiated cancellation workflow when a warehouse reports zero available-to-promise (ATP) quantity for a line item after order confirmation but prior to packing, notifying the customer with the reason for cancellation.
- **FR-011** The system shall record the cancellation reason (customer-initiated, stock shortfall, fraud check failure, address undeliverable) against every cancelled order for reporting purposes.

Splitting & Fulfillment

- **FR-012** The system shall evaluate order fulfillability at the SKU level against IMS and NetSuite WMS stock availability at each eligible warehouse before confirming an order.
- **FR-013** The system shall automatically split a multi-item order into two or more fulfillment segments when the full order quantity cannot be sourced from a single warehouse, assigning each segment an independent sub-order identifier linked to the parent order.
- **FR-014** The system shall independently track lifecycle status for each order segment while presenting a consolidated, aggregated order status to the customer on order tracking pages.
- **FR-015** The system shall recalculate shipping cost and estimated delivery date per order segment when a split occurs, and reflect any changes transparently to the customer.

Logistics Handoff

- **FR-016** The system shall select the appropriate logistics partner for each order segment based on destination country, with SwiftShip Logistics used for domestic India shipments and Continental Freight Partners used for UAE and Southeast Asia cross-border shipments.
- **FR-017** The system shall generate and transmit a shipment creation request, including consignee details, package weight/dimensions, and SKU manifest, to the assigned logistics partner's API upon order packing completion.
- **FR-018** The system shall ingest carrier tracking webhook events and reconcile them against the corresponding order segment within 5 minutes of receipt.
- **FR-019** The system shall allow authorized Order Operations users to manually reassign a logistics partner for an order segment in the event of a carrier service disruption.
- **FR-020** The system shall capture a customer-initiated return request against a delivered order and hand it off to the Returns & Refunds module with the associated order and item context, including an RMA reference number.

8. Non-Functional Requirements

Performance

- **NFR-001** Order status API calls shall respond within 2 seconds at the 95th percentile under normal load.
- **NFR-002** Order splitting evaluation logic shall complete within 3 seconds of order confirmation for orders containing up to 20 line items.

Security

- **NFR-003** The Order Management module shall comply with OWASP Top 10 security controls for all customer-facing and partner-facing APIs.
- **NFR-004** All order data containing personally identifiable information (PII), including shipping address and contact details, shall be encrypted using AES-256 at rest and TLS 1.2 or higher in transit.

Availability

- **NFR-005** The order status and order placement services shall maintain 99.9% uptime measured monthly.
- **NFR-006** Planned maintenance windows shall not overlap with declared peak sale events.

Scalability

- **NFR-007** The module shall support up to 10x normal order volume during peak sale events such as Diwali and Black Friday without degradation of order-to-ship SLA performance.
- **NFR-008** The order splitting and fulfillment evaluation engine shall scale horizontally to process a minimum of 5,000 order evaluations per minute during peak load.

Reliability

- **NFR-009** The Order Management module shall maintain a Recovery Point Objective (RPO) of 15 minutes and a Recovery Time Objective (RTO) of 1 hour for critical order processing services.
- **NFR-010** Order status transitions shall be idempotent and durably logged to prevent duplicate or lost state changes during system failover.

9. User Stories

- **US-001** As a customer, I want to see a single accurate order status regardless of which channel I check, so that I trust the delivery information provided.
- **US-002** As a customer, I want to cancel my order before it is packed, so that I am not charged or shipped an item I no longer need.
- **US-003** As a customer, I want to be notified immediately if an item in my order becomes unavailable, so that I can decide whether to wait, substitute, or cancel.
- **US-004** As an Order Operations Analyst, I want the system to automatically split orders across warehouses, so that customers receive available items without manual coordination delays.
- **US-005** As a Customer Service Representative, I want complete order history and status visible in Salesforce CRM, so that I can resolve customer inquiries without switching systems.
- **US-006** As a DevOps Lead, I want order status transitions logged and monitored in real time, so that I can detect and respond to processing failures quickly.
- **US-007** As a Product Owner, I want visibility into order-to-ship SLA compliance, so that I can prioritize fulfillment improvements.
- **US-008** As a Compliance Officer, I want order data encrypted and access-logged, so that ShopSphere meets its regulatory and data protection obligations.

10. Acceptance Criteria

AC ID	Related ID	Criteria	Expected Result
AC-001	FR-001	Given a customer completes checkout with successful payment, when the order is submitted, then an order record is created in "Placed" status within 2 seconds.	Order visible in "Placed" status on order history page.
AC-002	FR-002	Given inventory is reserved successfully in IMS, when reservation is confirmed, then the order status updates to "Confirmed".	Order status reflects "Confirmed" across all channels.

AC-003	FR-013	Given an order contains items available only across two warehouses, when the order is confirmed, then the system creates two linked sub-orders.	Two order segments created, each with independent tracking, linked to one parent order ID.
AC-004	FR-008	Given an order is in "Confirmed" status and not yet packed, when the customer requests cancellation, then the cancellation is accepted and inventory released.	Order status changes to "Cancelled"; stock reservation released in IMS.
AC-005	FR-010	Given a warehouse reports zero ATP after order confirmation, when the system detects the shortfall, then a system-initiated cancellation is triggered with customer notification.	Customer receives email/SMS explaining the cancellation reason within 15 minutes.
AC-006	FR-017	Given an order segment reaches "Packed" status, when packing is confirmed by NetSuite WMS, then a shipment creation request is sent to the assigned logistics partner.	Shipment created with tracking number returned within 10 minutes.
AC-007	FR-018	Given a carrier sends a tracking webhook update, when the event is received, then the corresponding order segment status is updated within 5 minutes.	Order status reflects latest carrier milestone (e.g., "Out for Delivery").
AC-008	US-005	Given a Customer Service Representative opens a case in Salesforce CRM, when they search by order ID, then complete order history and current status are displayed.	Full order timeline visible within Salesforce CRM without switching systems.
AC-009	FR-020	Given an order is in "Delivered" status, when a customer initiates a return, then an RMA reference is generated and handed off to the Returns & Refunds module.	Return request appears in Returns & Refunds queue with order context intact.

11. Business Rules

- **RULE-001** An order cannot be cancelled by the customer once it has entered "Packed" status; cancellation requests after this point must be routed to the Returns & Refunds module post-delivery.
- **RULE-002** An order must not transition to "Confirmed" status unless inventory has been successfully reserved for all line items or the affected items have been removed via order splitting.
- **RULE-003** Order splitting shall only occur across warehouses within the same fulfillment region (e.g., an India order shall not be split to fulfill from a UAE warehouse).
- **RULE-004** COD (Cash on Delivery) orders exceeding a value threshold defined by Finance shall require additional address and KYC verification before confirmation.
- **RULE-005** The order-to-ship SLA is defined as 48 hours from order confirmation for domestic (India) orders and 96 hours for cross-border (UAE/Southeast Asia) orders.
- **RULE-006** An order shall automatically move to "Closed" status 7 calendar days after "Delivered" status if no RMA has been initiated within that window.
- **RULE-007** System-initiated cancellations due to stock shortfall shall trigger an automatic full refund initiation for any pre-paid amount, coordinated with the Payments module.
- **RULE-008** Logistics partner assignment shall default to SwiftShip Logistics for all domestic India shipments and Continental Freight Partners for all UAE and Southeast Asia cross-border shipments, unless manually overridden by an authorized Order Operations user.

12. Process Flow Description

Primary Flow — Standard Order Lifecycle:

1. Customer completes checkout on the ShopSphere web or mobile app; payment is authorized via the PayEdge Payment Gateway (or COD is selected and validated).
2. Order Management creates the order record in "Placed" status and requests inventory reservation from the IMS and NetSuite WMS.
3. Upon successful reservation confirmation for all line items, the order transitions to "Confirmed" status and is synchronized to SAP S/4HANA for order-to-cash processing.
4. The Notification Service sends an order confirmation email (SendGrid) and SMS (Twilio) to the customer.
5. NetSuite WMS performs picking and packing; upon completion, a packing-complete event updates order status to "Packed".
6. Order Management generates a shipment creation request and hands it off to the assigned logistics partner (SwiftShip Logistics or Continental Freight Partners).

7. Upon receiving a shipment manifest confirmation, the order transitions to "Shipped" with tracking details captured and shared with the customer.
8. Carrier tracking webhooks progressively update order status to "Out for Delivery" and then "Delivered".
9. Seven calendar days after delivery, absent a return request, the order automatically transitions to "Closed".

Alternate/Exception Flow — Warehouse Stock Shortfall Triggering Split or Cancellation:

1. After order confirmation but prior to packing, NetSuite WMS or IMS reports that one or more line items have insufficient stock at the originally assigned warehouse.
2. Order Management re-evaluates fulfillability across alternate warehouses within the same fulfillment region.
3. If an alternate warehouse can fulfill the affected item(s), the system automatically creates a new order segment (split) linked to the parent order, and both segments proceed independently through packing and shipment.
4. If no warehouse within the region can fulfill the affected item(s), the system triggers a system-initiated cancellation for the affected line item(s), releases any reserved inventory, and initiates a refund workflow for the corresponding amount.
5. The customer is notified via email and SMS of the split (with updated delivery estimates) or the cancellation (with reason and refund timeline).
6. The event is logged for Order Operations reporting to track stock-shortfall frequency by SKU and warehouse.

13. Assumptions

- The Inventory Management System (IMS) and NetSuite WMS provide real-time or near-real-time stock availability data via API.
- SwiftShip Logistics and Continental Freight Partners expose stable, documented APIs for shipment creation and tracking webhooks.
- Customers have valid, verified contact information (email and mobile number) captured at account creation for notification delivery.
- SAP S/4HANA is available to receive order synchronization events without significant latency during standard business hours.
- Order Operations staff will be trained on manual override capabilities (e.g., logistics reassignment) prior to go-live.
- Peak sale event traffic patterns are reasonably predictable based on historical data from prior fiscal years.

14. Constraints

- The module must integrate with existing SAP S/4HANA, Salesforce CRM, and NetSuite WMS instances without requiring their replacement.
- Delivery timelines are constrained by third-party logistics partner capacity and service-level commitments outside ShopSphere's direct control.
- Regulatory requirements in India, UAE, and Southeast Asia impose differing KYC and COD handling rules that the system must accommodate per region.
- The program budget and FY2026 timeline require phased rollout, with domestic (India) order splitting prioritized ahead of cross-border capability.
- Legacy order data migration must preserve historical order references used by existing customer service cases in Salesforce CRM.

15. Dependencies

- Dependency on the Inventory Management System (IMS) and NetSuite WMS for real-time stock availability and reservation status.
- Dependency on the PayEdge Payment Gateway for payment authorization status prior to order creation.
- Dependency on SAP S/4HANA for order synchronization, financial posting, and revenue recognition.
- Dependency on SwiftShip Logistics and Continental Freight Partners APIs for shipment creation and tracking updates.
- Dependency on the Notification Service (SendGrid Email API and Twilio SMS Gateway) for customer communications.
- Dependency on the Returns & Refunds module to accept handoff of return-initiation requests.

16. Risks

Risk ID	Description	Likelihood	Impact	Mitigation
RISK-001	Inventory data lag between IMS/NetSuite WMS and Order Management leads to overselling and post-order cancellations.	Medium	High	Implement near-real-time stock sync with reservation locks and periodic reconciliation jobs.
RISK-002	Logistics partner API outages delay shipment creation and status updates.	Medium	High	Implement retry queues, partner failover options, and manual reassignment capability (FR-019).
RISK-003	Order splitting logic incorrectly assigns items across regions, causing customs or cross-border compliance issues.	Low	High	Enforce RULE-003 region-locking and add automated validation checks before segment creation.
RISK-004	Peak sale event traffic (10x normal load) causes order processing latency or downtime.	Medium	High	Load testing ahead of peak events and horizontal auto-scaling of the fulfillment evaluation engine.
RISK-005	Inconsistent order status shown across channels due to synchronization delays between microservices.	Medium	Medium	Establish a single order-status event bus with guaranteed delivery to all consuming channels.
RISK-006	PII exposure through improperly secured order/shipping data in transit or at rest.	Low	High	Enforce AES-256 encryption at rest, TLS 1.2+ in transit, and periodic security audits led by Compliance & Security.

17. Data Requirements

The Order Management module captures and maintains order-related data throughout the order lifecycle, feeding downstream systems including SAP S/4HANA, Salesforce CRM, and Tableau/Google Analytics 4 for reporting. Order records, including status history, shall be retained in the operational database for a minimum of 3 years to support customer service inquiries, audit requirements, and regulatory obligations, with archival storage thereafter per ShopSphere's enterprise data retention policy.

Data Field	Description
Order ID	Unique identifier for the parent customer order.
Sub-Order/Segment ID	Identifier for a split fulfillment segment linked to the parent Order ID.
Order Status	Current lifecycle state (Placed, Confirmed, Packed, Shipped, Out for Delivery, Delivered, Closed, Cancelled).
SKU List	List of product SKUs, quantities, and unit prices included in the order or segment.
Warehouse ID	Identifier of the fulfilling warehouse for each order segment.
Customer ID	Reference to the customer profile, including shipping address and contact details (PII).
Logistics Partner & Tracking Number	Assigned carrier (SwiftShip Logistics/Continental Freight Partners) and shipment tracking reference.
Cancellation Reason Code	Reason for order or line-item cancellation, where applicable.
Status Timestamp History	Timestamped log of every lifecycle status transition for SLA measurement.
RMA Reference	Return Merchandise Authorization number generated upon return initiation handoff.

18. Integration Requirements

System	Purpose	Direction	Data Exchanged
SAP S/4HANA	Order synchronization and revenue recognition	Outbound (Order Mgmt → SAP)	Confirmed order details, order value, cancellation/refund events
NetSuite WMS / IMS	Inventory reservation, stock availability, and packing status	Bidirectional	Stock reservation requests, ATP status, packing-complete events
SwiftShip Logistics & Continental Freight Partners	Shipment creation and tracking updates	Bidirectional	Shipment manifest data, tracking numbers, delivery status webhooks
Salesforce CRM	Order history and status visibility for customer service	Outbound (Order Mgmt → CRM)	Order timeline, status, cancellation/return context

Notification Service (SendGrid/Twilio)	Customer-facing status notifications	Outbound (Order Mgmt → Notification Service)	Order status change events, customer contact details
PayEdge Payment Gateway	Payment authorization confirmation prior to order creation	Inbound (Payments → Order Mgmt)	Payment authorization status, transaction reference

19. Reporting Requirements

- Real-time order status dashboard showing volume by lifecycle stage, segmented by region and warehouse.
- Order-to-ship SLA compliance report, broken down by domestic vs. cross-border shipments.
- Order cancellation report by reason code (customer-initiated, stock shortfall, fraud, undeliverable address), trended over time.
- Order splitting frequency report by SKU and warehouse to identify recurring inventory shortfall patterns.
- Logistics partner performance report, including shipment creation turnaround and on-time delivery rate.
- Executive summary dashboard in Tableau integrating order volume, GMV impact of cancellations, and SLA adherence trends for Rajesh Mehta and the E-Commerce leadership team.

20. Success Metrics

KPI	Target
Order-to-ship SLA compliance	≥ 95% of domestic orders shipped within 48 hours; ≥ 90% of cross-border orders shipped within 96 hours
Post-confirmation stock shortfall cancellation rate	≤ 1.5% of confirmed orders
Order status data consistency across channels	≥ 99.5% match rate between web, mobile, and CRM views
Automated order split success rate	≥ 98% of eligible split orders processed without manual intervention
Customer support tickets related to order status discrepancy	Reduction of 40% within 6 months post-launch
Shipment creation turnaround (packed to carrier handoff)	≤ 10 minutes at 95th percentile

21. Future Enhancements

- Predictive stock-shortfall detection using machine learning to preemptively suggest order splits before confirmation.
- Dynamic logistics partner selection based on real-time carrier performance and cost optimization rather than static regional rules.
- Customer self-service order modification (e.g., address change, delivery date preference) prior to packing.
- Integration with additional regional logistics partners to expand Southeast Asia last-mile coverage.
- Proactive customer communication enhancements, including WhatsApp-based order status notifications.

22. Glossary

Term	Definition
SKU	Stock Keeping Unit; a unique identifier for a distinct sellable product/variant.
PDP	Product Detail Page; the storefront page displaying detailed product information.
PLP	Product Listing Page; a storefront page displaying a list/grid of products.
COD	Cash on Delivery; a payment method where the customer pays at the time of delivery.
RMA	Return Merchandise Authorization; a reference issued to authorize and track a product return.
SLA	Service Level Agreement; a defined, measurable performance commitment.
ATP	Available to Promise; the quantity of stock available for new customer orders.
OTP	One-Time Password; a single-use code used for verification purposes.
KYC	Know Your Customer; identity verification process required for certain transactions.
GMV	Gross Merchandise Value; the total value of goods sold through the platform.

PII	Personally Identifiable Information; data that can identify a specific individual.
Order Segment	A sub-portion of a parent order created when an order is split across multiple warehouses.

23. Appendix

Reference Documents:

- ShopSphere Platform Modernization Program Charter, FY2026.
- BRD-EC-PAY-002 — Payments Module (PayEdge Payment Gateway integration).
- BRD-EC-RTN-006 — Returns & Refunds Module.
- BRD-EC-INV-004 — Inventory Management Module.
- ShopSphere Enterprise Data Retention Policy, v3.2.

Revision History:

Version	Date	Author	Change Description
0.5	20-Jan-2026	Neha Kapoor	Initial draft covering order lifecycle and cancellation requirements.
0.8	05-Feb-2026	Neha Kapoor	Added order splitting logic, logistics handoff, and NFR sections following architecture review with Sarah Chen.
0.9	12-Feb-2026	Neha Kapoor	Incorporated QA and Compliance feedback from Michael Torres and Fatima Al-Sayed; refined acceptance criteria and risk register.